COURSE TITLE : INTERIOR DESIGN ENGINEERING
COURSE CODE : 5121
COURSE CATEGORY : A
PERIODS PER WEEK : 4
SEMESTER : 5
PERIODS PER SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

Module	**Topics**	**Periods**
I	History of Interior Design-Fundamentals of Design Ergonomics Test	6 6 1
II	Interior Design Materials Construction Techniques Test	6 6 1
III	Designing of Residential spaces Colour Theory Test	10 2 1
IV	Designing of Public Spaces Lights and Luminaries Green Building Technology Test	9 3 1
	Total	**52**

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand fundamentals of interior designing
	2	Understand importance of ergonomics
2	1	Understand interior decoration materials
	2	Understand construction techniques
3	1	Understand interior designing of residential spaces
	2	Understand window treatment, wall treatment and floor treatment
	3	Know color theory
4	1	Understand interior designing of public spaces
	2	Understand Light and Luminaries
	3	Understand green building technology

Specific outcome

MODULE-I

1.1.0 Understand Fundamentals Of Interior Designing

- **1.1.1** Define Interior Design
- **1.1.2** State the Importance of Interior Design and decoration
- **1.1.3** Describe the History of Interior Design
- **1.1.4** Discuss fundamentals of designing
- **1.1.5** Discuss harmony proportion balance rhythm and emphasis

1.2.0 Understand Importance of Ergonomics

- 1.2.1 Define ergonomics
- 1.2.2 Explain Approach to human factor
- 1.2.3 Explain system reliability
- 1.2.4 Explain system reliability
- 1.2.5 Define Display
- 1.2.6 Explain Display with figures
- 1.2.7 Explain selection of Display
- 1.2.8 Define Graphic relationship
- 1.2.9 Explain Qualitative, Quantities reading and Check display
- 1.2.10 Explain Anthropometry

MODULE-II

2.1.0 Understand Interior Decoration Materials

- 2.1.1 List Construction Materials
- 2.1.2 Describe Stone, Brick and Timber
- 2.1.3 Classify cement and Explain Manufacture
- 2.1.4 Describe Concrete and Mortar
- 2.1.5 Explain wood based Construction Materials
- 2.1.6 Describe Glass, Metal, Paint, Plastic, rubber, latex ,Asphalt, Bitumen

2.2.0 Understand Construction Techniques

- 2.2.1 List Construction Techniques
- 2.2.2 Describe deep and shallow foundation with sketches
- 2.2.3 Explain Brickbonds
- 2.2.4 Explain arches and lintels

MODULE-III

3.1.0 Understand Interior Designing Of Residential Spaces

- 3.1.1 State the Importance Of Residential Spaces
- 3.1.2 Describe Minimum requirements of Residential Spaces
- 3.1.3 Explain Design Consideration of Living room, Dining room, Kitchen, Bedroom, Study room, Bathroom, Library, Storage room with Figure

3.2.0 Understand Window Tratment, Wall Treatment And Floor Treatment

3.2.1 Explain wall treatment, Window treatment, Floor treatment
3.2.2 Describe Carpet and Cushion
3.2.3 Explain Design sinages and symbols
3.2.4 Explain Furniture Arrangement with minimum Dimensions

3.3.0 Know Colour Theory

3.3.1 State colour Theory
3.3.2 Illustrate Colourwheel and Explain
3.3.3 Describe the Psychology of Colour Theory
3.3.4 Explain Air conditioning Fundamentals

MODULE-IV

4.1.0 Understand Interior Designing of Public Spaces

4.1.1 Describe designing of office interiors and drawing with examples
4.1.2 Describe the designing of general office multiple work station private Offices and reception
4.1.3 Describe hospitality spaces such as hotels restaurants
4.1.4 Describe the plant scaping a audio visual system auditorium sealing security
4.1.5 Describe the colour theory window treatment (general idea only)
4.1.6 Understand designing a room
4.1.7 Describe floor plans and room arrangement, Dimensions
4.1.8 Describe decorating a room
4.1.9 Describe the selection of color fabrics, wall treatment, floor coverings, window treatment, lighting and accessories
4.1.10 Compute the estimate and budget for public spaces such as seminar hall theatre

4.2.0 Understand Light and Luminaries

4.2.1 Define Light
4.2.2 Describe Different Light Sources
4.2.3 State Luminaries
4.2.4 Classify Luminaries

4.3.0 Understand Greenbuilding Technology

4.3.1 State Green Building
4.3.2 Describe the relevance of Greenbuliding
4.3.3 Explain Greenbuilding materials
4.3.4 Describe the Fundamentals of Estimating and Costing
4.3.5 Calculate Sample Estimations

<u>CONTENT DETAILS</u>

MODULE-I

Development of interior design- concepts - historic review. Place of interior in the modern era- changing trends and salient features, objectives of aesthetic planning - Beauty, expressiveness, functionalism, economy Role of good taste – meaning and importance. Need for developing skill in aesthetics Elements

of design - Line and direction, form and shape, size, colour, light, pattern, texture and space - application of elements to form designs.
Principles of design –Balance, rhythm, emphasis, harmony, proportion - meaning and application of design concepts in the interior and exterior houses and other commercial buildings.

Development of Ergonomics – Objectives – Approach to human factors – system- man machine system – system reliability – Display –Type of Display – selection of Display —compatibility Process of seeing - visual – visual compatibilities – Graphic representation it data –symbols – Quantitative reading – Quanlitative reading – Quantitative reading –visual Display –check Display –Work Place Design Anthropometry –static Dimensions – Dynamic Dimensions – Use of Anthropometric data.

MODULE-II

Building materials and finishes - Types and uses of stone, brick, timber, cement, mortar, concrete, plastics, glass, wood based materials, metals - ferrous and nonferrous, wall, floor and ceiling finishes.

various construction techniques in interiors. Foundation –Deep and Shallow, Brick bonds - stretcher, header, English and Flemish bond , Arches, Lintels, Staircase, cladding, flooring, roofing, ceiling

MODULE-III

Planning and designing of interior spaces- Furniture dimensions – standard dimension of furniture planning & design of interior spaces minimum requirements- planning considerations and drawings preparation of dining room, bed room, kitchen, bath room, library study room, storage area Planning and designing of interior spaces –- design sinages and symbols. Free hand drawings

Concept of colour - significance of colour in the interiors and exteriors-Dimensions of colour –Hue, value, intensity, Effects of Hue, value and Intensity.
Colour systems - -planning colour harmonies-related and contrast. Factors considered in selecting colour harmonies. –Colour wheel-Colour theory.
Application of colour harmonies in the interiors and exteriors –Effects of light on colour, Illusion of colour, psychology of colour, effect of colour on each other

MODULE-IV

General office multiple workstation-private offices -reception -hospitality spaces restaurants- hotel public spaces- private offices reception hospitality spaces- restaurants –hotel- public spaces – seminar hall- theatre- Auditorium- wall finishes floor and floor finishes-Wall treatment-Window Treatment and Floor Finishes.

Importance of lighting – Artificial lighting - light sources,Types and uses of light, specific factors in lighting – measurements of lighting and economy in lighting, Psychological aspects of light, Avoidance of glare – Glare its types and prevention.

Lighting accessories – Selection of lamps and lighting fixtures, lighting for various areas and specific activities, modern features in lighting design.

Green building technology – Meaning, concept, impact of green building on human health and natural environment, need, importance and benefits of green buildings.
Materials and finishes used in green building – Bamboo, straw, wood, dimension stone, Recycled stone, non-toxic metals, Earth blocks-compressed, rammed, baked; vermiculites, flax linen, sisal, wood fibres, cork, coconut ,polyurethane block.
Green building practices and technologies. Roof, walls, floors – electrical, plumbing, windows, and doors, heating, ventilation and air conditioning (HVAC), insulation, Interior

REFERENCE BOOKS

1. John A Walton, Wood works in theory and practice –Australian publishing co.

2. Carpentry a complete guide – McGraw Hill, New Delhi.

3. Eric Stephensor and Dave plank, Rochdati, Circular Saw –Thom Robinsons and Son Ltd. England.

4. Robert Schalff ,Complete Book of Wood finishing –

5. Ernest Scott, Working in wood –

6 Joseph De Chiala, Julius Pauero. Time Saver standards for interior Design and –, Space planning Martin Zelink, McGraw Hill, New York.

7. Pratap R.M (1988), Interior design principles and practice, Standard publisher's distribution, Delhi.

8. Rangawala, S.C Engineering Materials, Charter publishing house,

9. Rangawala, S.C, Building construction, Charter publishing house,

10. Rangawala .S.C., Water supply and sanitary Engineering, publishing house, Roorkee

11. Seetharam, P and Pannu, P.Interior Design and Decoration, CBS publishers and distributors, NewDelhi

12. P.C Sharma, Production Technology -

13. Ahamed A. Kasu ,An introduction to Art and Craft Science - Ashish Book Cen. Mumbai

14. Phyllis Solan Allen Lynn . M. Jones Mirian F Stimpns ,Beginnings of Interior Environment - Pearson Prentice hall

15. Planning Designing and Decorating a room, Step by Step - Charlotte Finn Person Prentice hall

16. Marthand Testland, Industrial Management and Production Engineering -

17. Mickomick, Human factors & Engineering